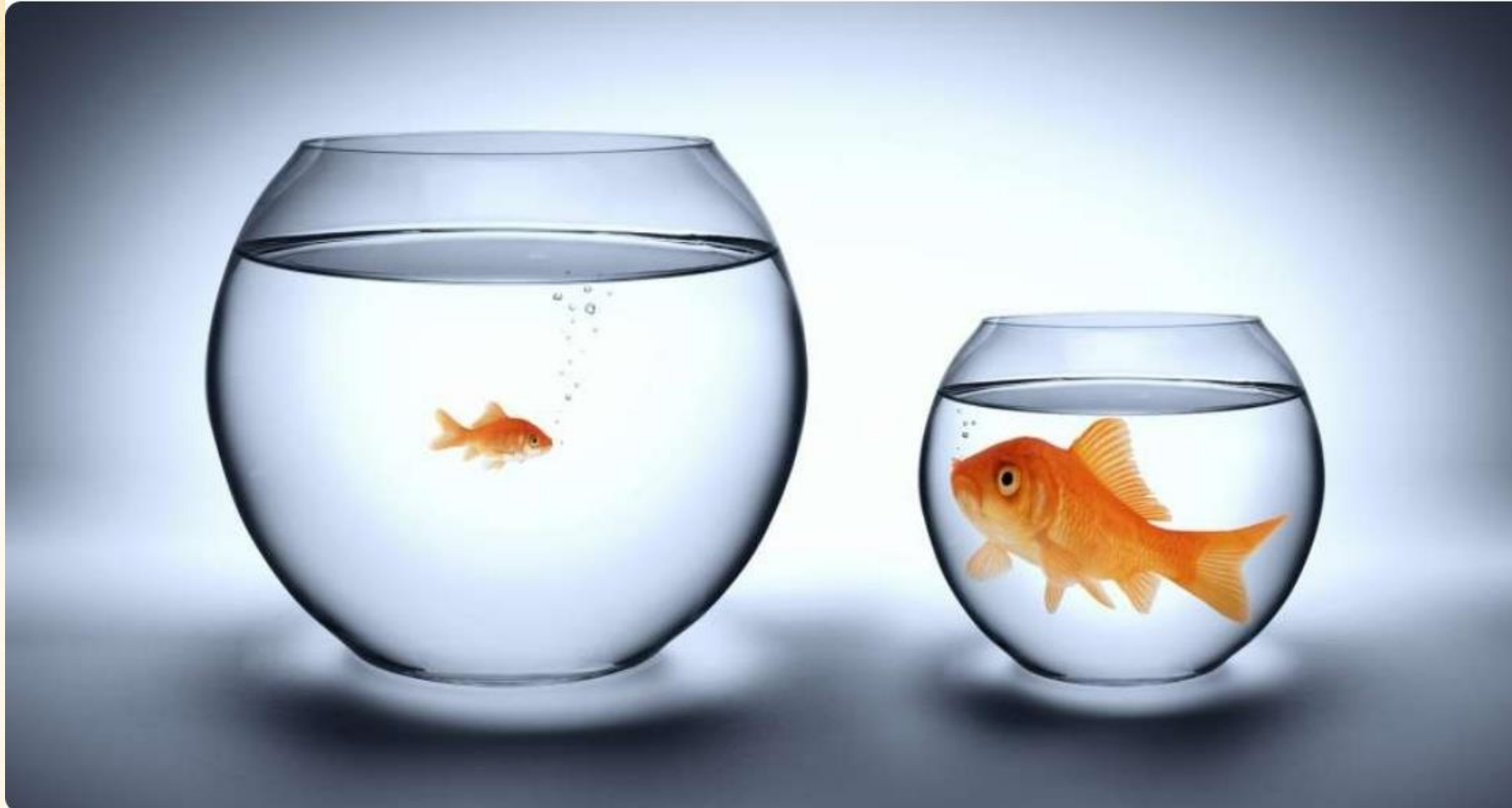


Big-Fish-Little-Pond Effect Research with TIMSS Data

- **Wang, Z.** (2020). When large-scale assessments meet data science: The big-fish-little-pond effect in fourth- and eighth-grade mathematics across nations. *Frontiers in Psychology*, 11(2427). <https://doi.org/10.3389/fpsyg.2020.579545>
- **Wang, Z., & Bergin, D.** (2017). Perceived relative standing and the big-fish-little-pond effect in 59 countries and regions: Analysis of TIMSS 2011 data. *Learning and Individual Differences*, 57, 141-156. doi: 10.1016/j.lindif.2017.04.003
- **Wang, Z.** (2015). Examining big-fish-little-pond-effects across 49 countries: a multilevel latent variable modelling approach. *Educational Psychology*, 35(2), 228-251. doi: 10.1080/01443410.2013.827155

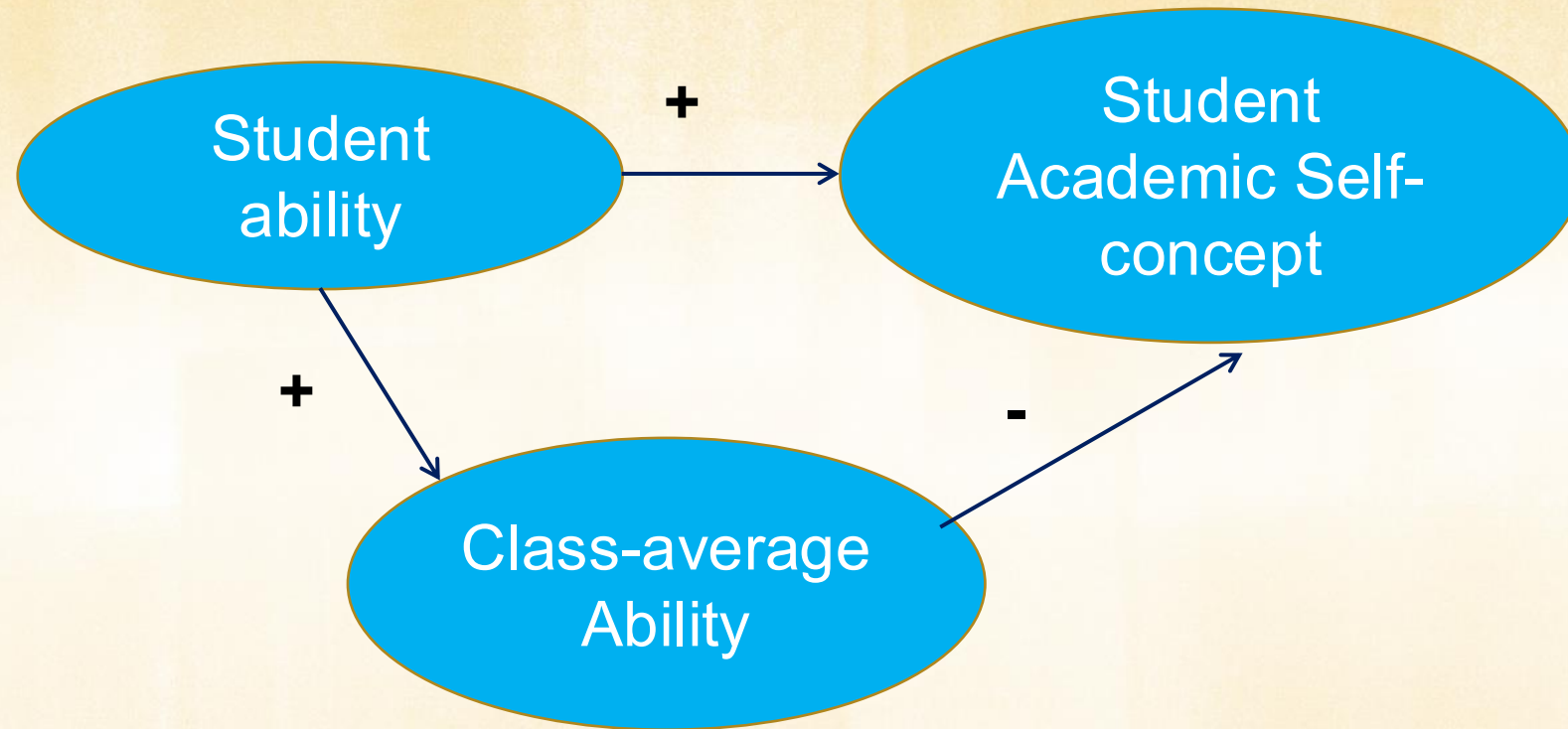
The Big-Fish-Little-Pond Effect



Picture source: <https://www.linkedin.com/pulse/small-fish-big-pond-gaurav-kumar/>

The Big-Fish-Little-Pond Effect

Model



TIMSS Data

- Complex data structure
 - Unequal selection probabilities
 - Clustering
- Matrix sampling technique
 - Use of IRT and multiple imputation to obtain plausible values to represent academic achievement
 - Five plausible values for each student

Statistical Model

- Multilevel latent covariate model (Lüdtke, et al., 2008) treats contextual effects using **implicit group mean centering**
 - Observed variables at level 1 are decomposed into unobserved components of group-specific deviations and individual deviations within group

$$X_{ij} = \mu_x + U_{xj} + R_{xij} \quad Y_{ij} = \mu_y + U_{yj} + R_{yij}$$

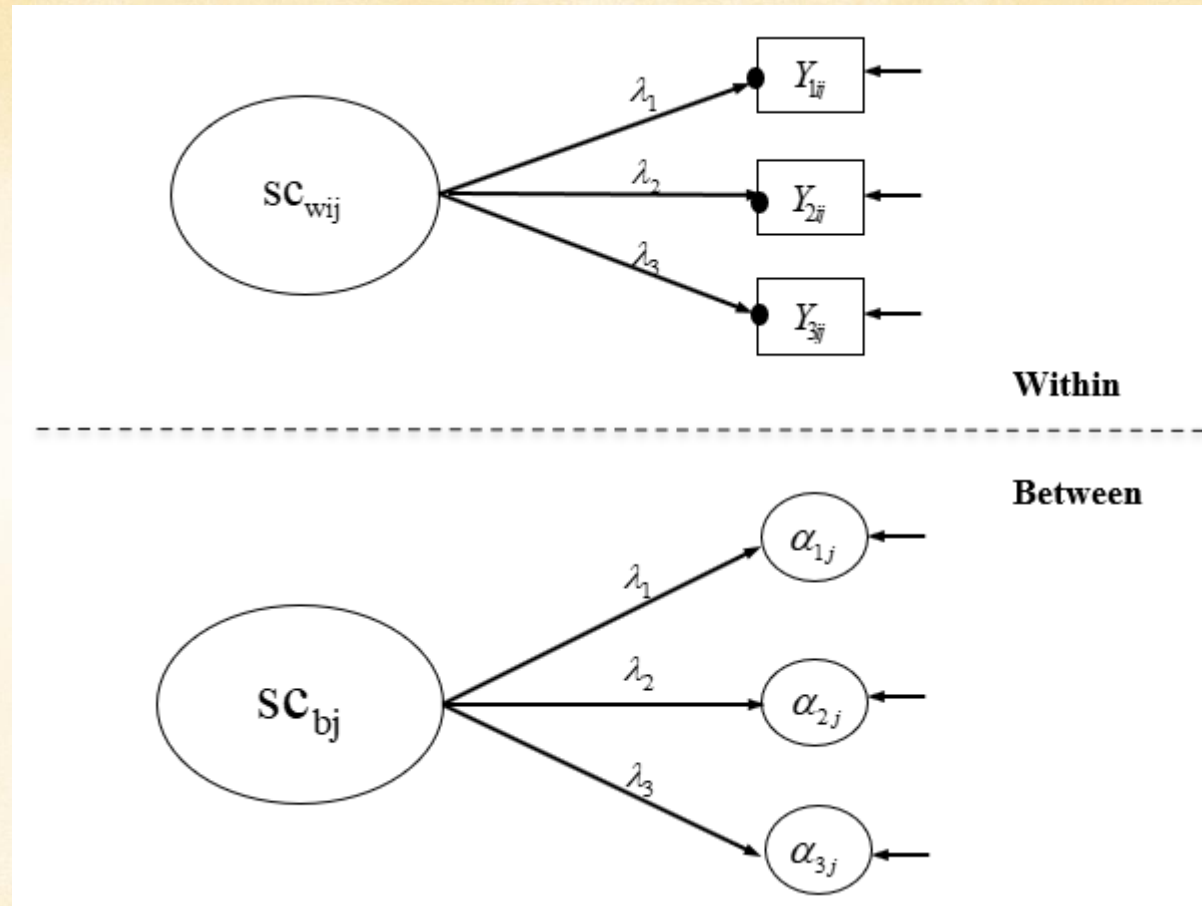
- Relationship between X and Y is decomposed into within- and between-level components

$$R_{yij} = \beta_{\text{within}} R_{xij} + \varepsilon_{ij} \quad U_{yj} = \beta_{\text{between}} U_{xj} + \delta_j$$

- BFLPE exists if $(\beta_{\text{within}} - \beta_{\text{between}})$ is negative and significant

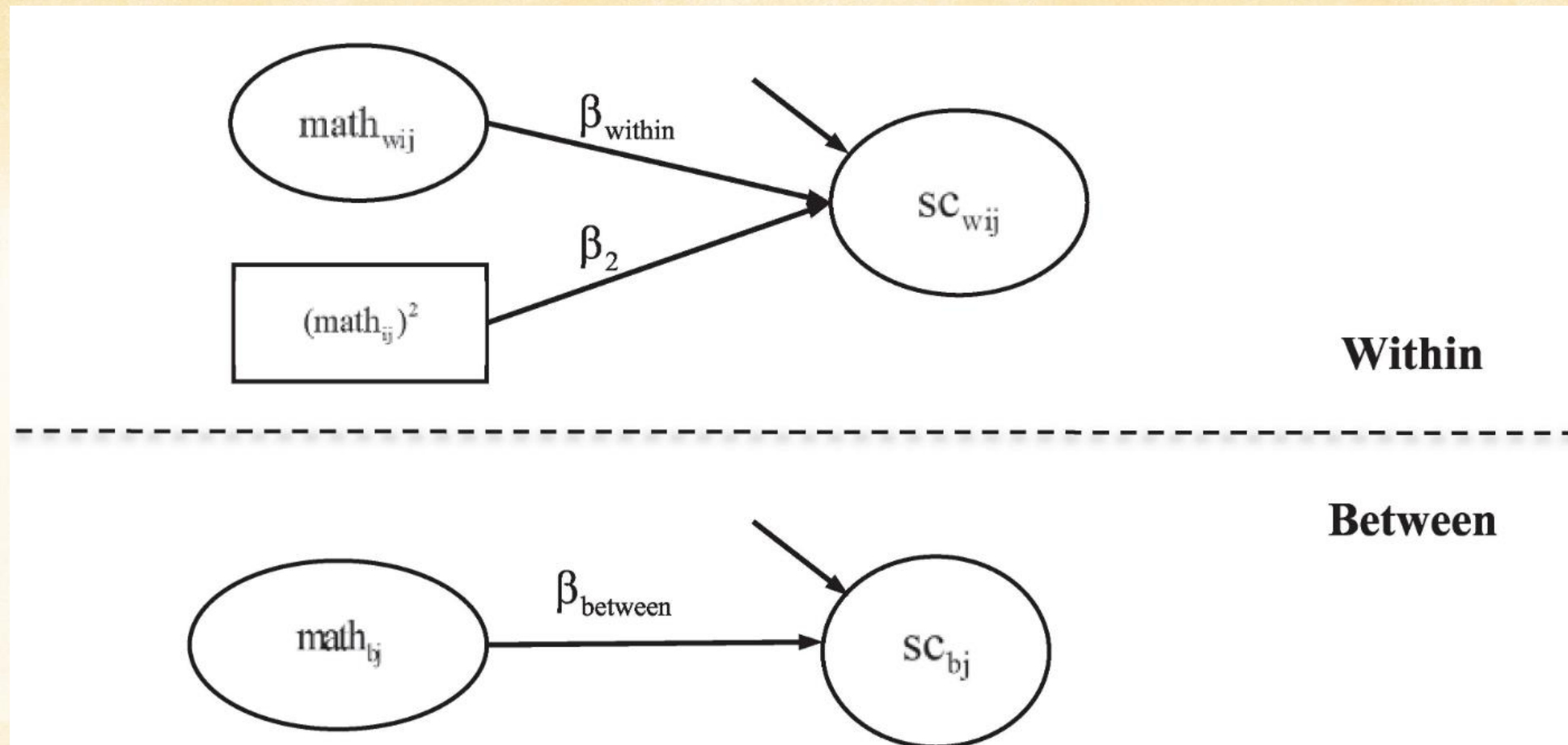
The Big-Fish-Little-Pond Effect

Model for Self-Concept with Multilevel Measurement Invariance



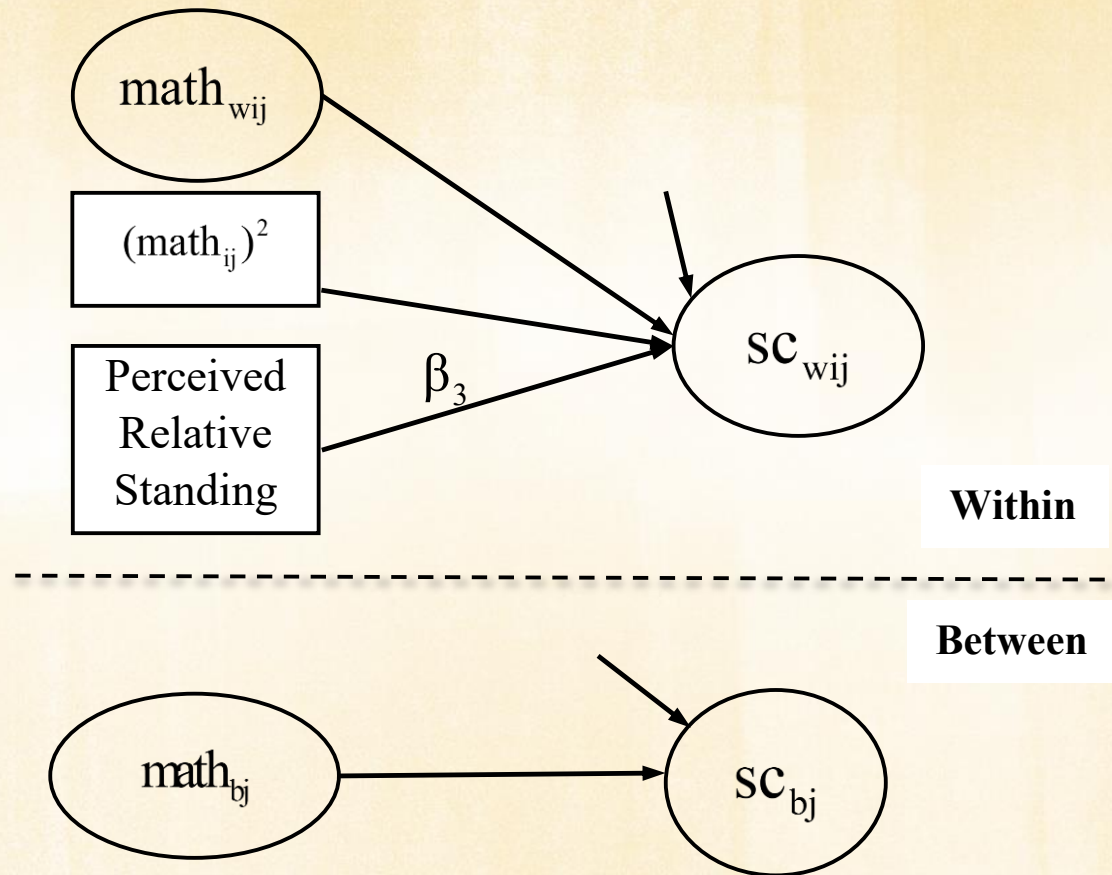
The Big-Fish-Little-Pond Effect

BFLPE Model



The Big-Fish-Little-Pond Effect

BFLPE with Perceived Relative Standing



General Findings Using Three Waves of TIMSS Data Collected in 2007, 2011, and 2015

- There is cross-level measurement invariance of self-concept in most countries/regions.
- Measurement invariance of math self-concept does not hold across countries/regions.
- There is generalizability of the BFLPE across countries and regions.
- Including Perceived Relative Standing reduces but typically does not eliminate the BFLPE.
- Large-scale analysis is now easier than before.
 - Use of R packages: EdSurvey, MplusAutomation, tidyverse